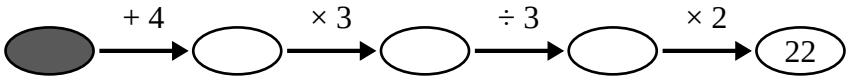
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 27966

Part A: Each question is worth 3 points

1. Mia wrote a favourite number in the dark cloud and did the calculations shown by the arrows, one after the other. The result was 22.



What is Mia's favourite number?

A. 9 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

2. How many common letters do the words MATHEMATICS and TRIANGLE have?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

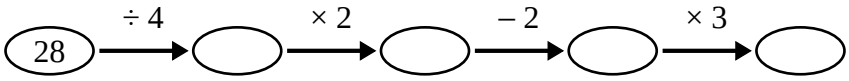
3. How many different letters are used to write the word BUTTERFLY?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

4. How many common letters do the words MATHEMATICS and CASTLE have?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 9

5. Follow the arrows and do each calculation in turn.



Which number comes out at the end?

A. 28 ***B.*** 35 ***C.*** 36 ***D.*** 37 ***E.*** 44

6. How many different letters are used to write the word CHOCOLATE?

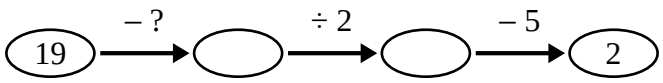
A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

Part B: Each question is worth 4 points

7. Emma thinks of a number, multiplies by 3, then divides by 3, then divides by 5, and gets 5. Which number did Emma think of?

A. 27 ***B.*** 25 ***C.*** 23 ***D.*** 5 ***E.*** 1

8. The arrows show calculations done one after the other. One number is hidden.



Which number is hidden behind the question mark?

A. 17 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 5

9. Mia writes the word TEACHER. Then Mia crosses out every letter that also appears in the word TRIANGLE.
How many different letters are **not** crossed out?

A. 4 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

10. Which word has exactly 2 different letters in common with DINOSAUR?

A. TURTLE ***B.*** PUZZLE ***C.*** KANGAROO ***D.*** WINDOW ***E.*** DRAGON

11. Leo writes the word LEMONADE. Then Leo crosses out every letter that also appears in the word SPIDER.
How many different letters are **not** crossed out?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 5

12. Emma thinks of a number, adds 7, then divides by 2, then adds 8, and gets 15.
Which number did Emma think of?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 15 ***E.*** 19

13. Emma thinks of a number, multiplies by 2, then adds 5, then multiplies by 3, and gets 39.
Which number did Emma think of?

A. 249 ***B.*** 39 ***C.*** 6 ***D.*** 5 ***E.*** 4

14. Mia writes the word TEACHER. Then Mia crosses out every letter that also appears in the word ELEPHANT.
How many different letters are **not** crossed out?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

15. Which word has exactly 1 different letter in common with COMPUTER?

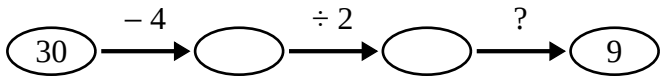
A. SANDWICH ***B.*** GIRAFFE ***C.*** ROCKET ***D.*** TEACHER ***E.*** GARDEN

Part C: Each question is worth 5 points

16. How many different letters are used to write the two words LEMONADE and NUMBER?

A. 9 ***B.*** 10 ***C.*** 11 ***D.*** 13 ***E.*** 14

17. The arrows show calculations done one after the other, but one arrow has lost its calculation: it shows a question mark.



Which calculation belongs on that arrow?

A. – 4 ***B.*** × 4 ***C.*** + 2 ***D.*** + 8 ***E.*** + 4

18. How many different letters are used to write the two words BANANA and DINOSAUR?

A. 10 ***B.*** 9 ***C.*** 8 ***D.*** 7 ***E.*** 6

19. How many different letters are used to write the two words WINDOW and SANDWICH?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 13

20. How many different letters are used to write the two words KANGAROO and WINDOW?

A. 8 ***B.*** 9 ***C.*** 11 ***D.*** 12 ***E.*** 14

Answer Key

1.	**B**	**2.**	**C**	**3.**	**A**	**4.**	**A**	**5.**	**C**	**6.**	**B**
7.	**B**	**8.**	**E**	**9.**	**C**	**10.**	**A**	**11.**	**E**	**12.**	**B**
13.	**E**	**14.**	**E**	**15.**	**A**	**16.**	**B**	**17.**	**A**	**18.**	**B**
19.	**D**	**20.**	**B**								

Solutions

1. B (7) — Go backwards and do the opposite: 22 ÷ 2 = 11, 11 × 3 = 33, 33 ÷ 3 = 11, 11 – 4 = 7. [set-03/by_level/Grade_03-04/2007 Q5 · favourite_number]

2. C (4) — Check each different letter of MATHEMATICS: only A, E, I and T are also in TRIANGLE, so 4. A letter that appears twice still counts once. [set-03/by_level/Grade_03-04/2007 Q3 · common_letters]

3. A (8) — The letters are B, E, F, L, R, T, U and Y: 8 different letters (BUTTERFLY has 9 letters in all). [set-03/by_level/Grade_03-04/2007 Q3 · different_letters]

4. A (5) — Check each different letter of MATHEMATICS: only A, C, E, S and T are also in CASTLE, so 5. A letter that appears twice still counts once. [set-03/by_level/Grade_03-04/2007 Q3 · common_letters]

5. C (36) — 28 ÷ 4 = 7, 7 × 2 = 14, 14 – 2 = 12, 12 × 3 = 36. [set-03/by_level/Grade_03-04/2007 Q5 · forwards]

6. B (7) — The letters are A, C, E, H, L, O and T: 7 different letters (CHOCOLATE has 9 letters in all). [set-03/by_level/Grade_03-04/2007 Q3 · different_letters]

7. B (25) — Go backwards: 5 × 5 = 25, 25 × 3 = 75, 75 ÷ 3 = 25. [set-03/by_level/Grade_03-04/2007 Q5 · think_of_a_number]

8. E (5) — Forwards from 19 we reach 19 before the hidden step; backwards from 2 we reach 14 after it. 19 – 5 = 14. [set-03/by_level/Grade_03-04/2007 Q5 · missing_number]

9. C (2) — Crossed out: A, E, R and T (they are in TRIANGLE). Left: C and H, so 2 different letters. [set-03/by_level/Grade_03-04/2007 Q3 · only_in_first]

10. A (TURTLE) — TURTLE shares R and U with DINOSAUR. PUZZLE shares 1; KANGAROO shares 4; WINDOW shares 4; DRAGON shares 5. [set-03/by_level/Grade_03-04/2007 Q3 · which_word]

11. E (5) — Crossed out: D and E (they are in SPIDER). Left: A, L, M, N and O, so 5 different letters. [set-03/by_level/Grade_03-04/2007 Q3 · only_in_first]

12. B (7) — Go backwards: 15 – 8 = 7, 7 × 2 = 14, 14 – 7 = 7. [set-03/by_level/Grade_03-04/2007 Q5 · think_of_a_number]

13. E (4) — Go backwards: 39 ÷ 3 = 13, 13 – 5 = 8, 8 ÷ 2 = 4. [set-03/by_level/Grade_03-04/2007 Q5 · think_of_a_number]

14. E (2) — Crossed out: A, E, H and T (they are in ELEPHANT). Left: C and R, so 2 different letters. [set-03/by_level/Grade_03-04/2007 Q3 · only_in_first]

15. A (SANDWICH) — SANDWICH shares C with COMPUTER. ROCKET shares 5; TEACHER shares 4; GIRAFFE shares 2; GARDEN shares 2. [set-03/by_level/Grade_03-04/2007 Q3 · which_word]

16. B (10) — LEMONADE uses 7 different letters and NUMBER uses 6. E, M and N are in both, so count them once: 7 + 6 – 3 = 10. [set-03/by_level/Grade_03-04/2007 Q3 · letter_cards]

17. A (– 4) — Before that arrow we have 13; working back from 9, after it we need 9. Only – 4 turns 13 into 9. [set-03/by_level/Grade_03-04/2007 Q5 · which_operation]

18. B (9) — BANANA uses 3 different letters and DINOSAUR uses 8. A and N are in both, so count them once: 3 + 8 – 2 = 9. [set-03/by_level/Grade_03-04/2007 Q3 · letter_cards]

19. D (9) — WINDOW uses 5 different letters and SANDWICH uses 8. D, I, N and W are in both, so count them once: 5 + 8 – 4 = 9. [set-03/by_level/Grade_03-04/2007 Q3 · letter_cards]

20. B (9) — KANGAROO uses 6 different letters and WINDOW uses 5. N and O are in both, so count them once: 6 + 5 – 2 = 9. [set-03/by_level/Grade_03-04/2007 Q3 · letter_cards]